

Thandeka Ngetu

Exercise 1: SQL fundamentals

1. SELECT statement

```
SELECT *  
FROM employees;
```

2. SELECT DISTINCT statement

```
SELECT DISTINCT (department) AS Department_Name  
FROM employees;
```

3. ORDER BY statement

```
SELECT first_name,  
       last_name,  
       salary  
FROM employees  
ORDER BY salary DESC;
```

last_name	last_name	Salary
Emily	Wilson	62 000
Mike	Doe	60 000
Robert	Brown	59 000
John	Clark	55 000
Sarah	White	53 000
Daniel	Moore	53 000
David	Hall	52 000
Jessica	Smith	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4. LIMIT statement

```
SELECT first_name,  
       last_name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 5;
```

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000

5. WHERE Statement

```
SELECT first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'IT';
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. AND statement

```
SELECT first_name,  
       last_name,  
       department,  
       salary
```

first_name	last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

```
FROM employees
```

```
WHERE department = 'Finance' AND salary > 58000;
```

7. OR statement

```
SELECT first_name,  
       last_name,  
       department
```

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

```
FROM employees
```

```
WHERE department = 'HR' OR department = 'Marketing';
```


8. NOT Statement

```

SELECT first_name,
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department <> 'IT';

```

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. IN Statement

```

SELECT first_name,
       last_name,
       department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');

```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

```

10. SELECT first_name,
          last_name,
          department,
          salary,
          city

```

first_name	last_name	dept	salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	55000	New York

```

FROM employees
WHERE department = 'IT' AND city = 'New York'
AND salary > 50000;

```


11. SELECT first_name,
last_name,
department,
salary
FROM employees
WHERE department = 'Finance' AND department = 'Marketing'
ORDER BY salary > 52000;

12. SELECT DISTINCT (city) AS City_Name,
department
FROM employees
WHERE department IN ('Finance', 'Marketing');

13. SELECT first_name,
last_name,
department,
salary,
hire_date
FROM employees
WHERE salary > 50000 AND department <> 'Finance'
ORDER BY hire_date ASC;

14. SELECT first-name,
last-name,
department,
city

FROM employees

WHERE city = 'Chicago' OR city = 'Los Angeles'
AND IN ('IT', 'Marketing')

LIMIT 3;

first-name	last-name	department	city
Jane	Smith	HR	Chicago
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago

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first-name	last-name	department	salary	hire-date
Emily	Davis	IT	62000	2018-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

⑭

City-Name	department
Los Angeles	Finance
San Francisco	Marketing
Houston	Finance
Chicago	Marketing